

Computational Thinking

What is Computational Thinking



Thinking like a Computer:
It involves breaking down problems into smaller parts, finding patterns, and designing solutions that can be executed by a computer

Problem-Solving Approach:
Computational thinking is a way of thinking about problems that enables us to solve complex challenges

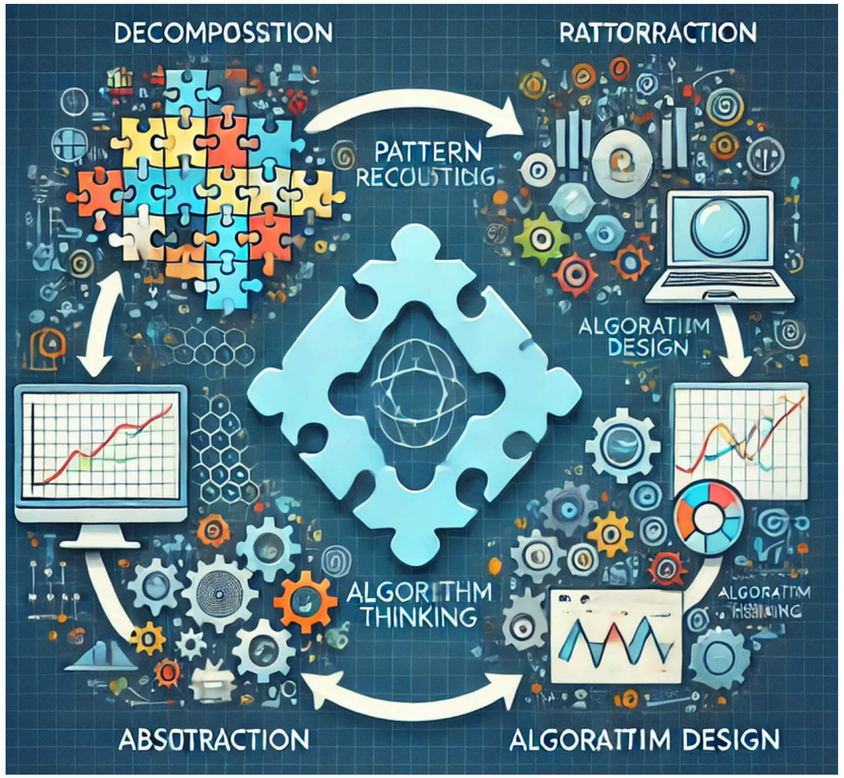
4 Key Skills of Computational Thinking

Decomposition
Breaking down a complex problem into smaller, manageable parts.

Pattern Recognition
Identifying similarities or patterns among the smaller sub-problems.

Abstraction
Removing unnecessary details to focus on the main idea or the most important aspects of a problem.

Algorithm Design
Creating a step-by-step procedure for solving the problem.



5 STAGES OF COMPUTATIONAL THINKING

Decomposition

Dividing a complex problem into smaller, easier-to-solve parts.



Generalisation & Pattern Recognition

Identifying patterns across different problems and applying them to solve new problems.



Modelling

Creating a simplified representation of a real-world system or problem.



Abstraction

Focusing on essential information while ignoring irrelevant details.



Evaluation

Assessing a solution to determine if it meets the desired outcomes and performance criteria.



DECOMPOSITION

Divide and Conquer

Decomposition helps simplify complex problems into smaller, manageable parts.

Break Down Complex Problems

Solve Each Part Independently

Focus on solving each smaller problem one at a time, making the task less daunting

Once each part is solved, the solutions are integrated to create a complete solution to the original problem.

Combine Solutions for Final Result

GENERALIZATION & PATTERN RECOGNITION

PATTERN RECOGNITION

Looking for trends or regularities within the problem. Often, different problems share commonalities that can be used to develop a generalized approach to problem-solving.

GENERALIZATION

Once a pattern is recognized, it can be generalized to apply to similar problems. This allows problem-solvers to avoid solving the same problem multiple times, improving efficiency

MODELLING

Visualizing the Problem

REPRESENT COMPLEX PROBLEMS

Modeling breaks down intricate problems into simpler, more manageable components, making them easier to analyze.



EASY TO UNDERSTAND

Visual models use intuitive representations, enabling even non-experts to grasp the core aspects of a problem quickly.



EASY TO IDENTIFY PATTERNS

By presenting data visually, modeling highlights recurring trends and correlations that might be missed in text or data.



MAKES AN IMPACT

Visual models deliver immediate insights, allowing viewers to grasp key points at a glance and make faster decisions.

ABSTRACTION: FOCUSING ON THE ESSENTIALS



SIMPLIFYING COMPLEXITY

Abstraction involves identifying the most relevant information while ignoring unnecessary details. It helps us focus on the core problem.

CREATING MODELS

We build simplified models that capture the essence of the problem without getting bogged down in every aspec

GENERALIZING CONCEPTS

Abstraction allows us to generalize patterns and apply them to different situations, making our solutions more broadly applicable.

Evaluation

Assessing and Refining Solutions



Test and Analyze

Run tests to identify weaknesses and areas for improvement.



Identify Errors

Analyze results to uncover any bugs or incorrect logic.



Refine and Iterate

Modify the solution based on findings to enhance accuracy and efficiency.

APPLYING COMPUTATIONAL THINKING

Where to apply?

EVERYDAY PROBLEM SOLVING

Computational thinking helps solve problems in everyday life, from planning a trip to managing finances.



SOFTWARE DEVELOPMENT

Programmers use computational thinking to design algorithms, build efficient code, and debug programs.



DATA ANALYSIS

Data analysts use computational thinking to identify patterns, draw insights, and make informed decisions.



RESEARCH AND INNOVATION

Scientists and researchers utilize computational thinking to model complex systems and design new technologies.

Case study: Designing a Traffic Management System.

Stage	Task	Output
Decomposition	Break down the traffic system into key components: roads, vehicles, signals, pedestrian crossings.	A list of manageable subsystems (e.g., traffic lights, sensors, vehicle types).
Generalisation & Pattern Recognition	Identify common patterns, such as peak traffic hours, accident hotspots, and traffic flow patterns.	Common traffic patterns and scenarios (e.g., rush hour, traffic congestion areas).
Modelling	Create a model to simulate traffic flow, delays, and signal timing using real-time data.	A dynamic simulation of the traffic system showing congestion and flow patterns.
Abstraction	Focus on key elements by omitting unnecessary details, such as only simulating main roads and intersections.	Simplified model focusing on key roads and high-traffic intersections.
Evaluation	Test the model by evaluating different traffic management strategies, such as signal timing or rerouting.	Performance metrics, such as reduced congestion, travel time, or improved safety.

Computational Thinking Alternative

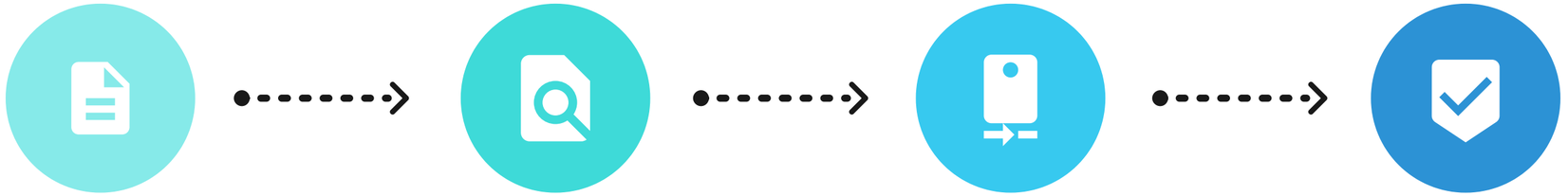


Logical Thinking:
Logical thinking refers to the ability to reason in a structured and clear manner, ensuring that conclusions follow logically from premises.

Algorithmic Thinking:
Algorithmic thinking involves breaking down complex problems into a series of smaller, more manageable steps.

LOGICAL THINKING: BREAKING DOWN PROBLEMS

Logical reasoning helps ensure that the steps taken in solving a problem make sense and lead to correct outcomes



1 - Identify the Problem

First, clearly define the problem you're trying to solve. What are the specific **goals** and constraints?

2 - Analyze the Problem

Break down the problem into smaller, manageable parts. Look for patterns, **relationships**, and potential solutions.

3 - Formulate a Solution

Develop a step-by-step plan to address each part of the problem. Ensure the steps are **logical** and interconnected.

4 - Evaluate the Solution

Test the solution to ensure it effectively addresses the problem. **Analyze** the results and make adjustments if needed.

ALGORITHMIC THINKING: STEP-BY-STEP SOLUTIONS

Algorithmic thinking involves breaking down complex problems into a series of smaller, more manageable steps. By defining a clear sequence of instructions, you can systematically solve problems, leading to efficient and reliable solutions.

Check if the solution meets the original goal.

5 - Evaluate Results

Follow the instructions in order to arrive at a solution.

4 - Execute the Steps

Order the steps logically to ensure a solution.

3 - Sequence Instructions

Break the problem into smaller steps to solve.

2 - Identify Steps

Clearly understand the problem you're trying to solve.

1 - Define the Problem



The Future of Computational Thinking

Computational thinking is rapidly evolving, driven by advancements in technology and its growing influence on various fields.

As technology progresses, so too will computational thinking, shaping the way we solve problems, learn, and interact with the world

THANK YOU
Q&A